

4. Basics of Probabilistic Modelling

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MTH345: Deep Generative Models: A Statistical Perspective



Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Statistical Estimation and Hypothesis Testing
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics

- ▶ We have previously formulated Bayesian networks and explored statistical inference based on them.
- ▶ In this chapter, we begin by reviewing two fundamental paradigms in probabilistic modeling: discriminative and generative models.
- ▶ After discussing their limitations in high-dimensional settings, we will transition into traditional dimension reduction methods.
- ▶ Recommended reference materials include:
 - ▶ Efron (1975), "The Efficiency of Logistic Regression to Normal Discriminant Analysis," *JASA*.
 - ▶ Section 14 in Hastie (2009), *The Elements of Statistical Learning*.

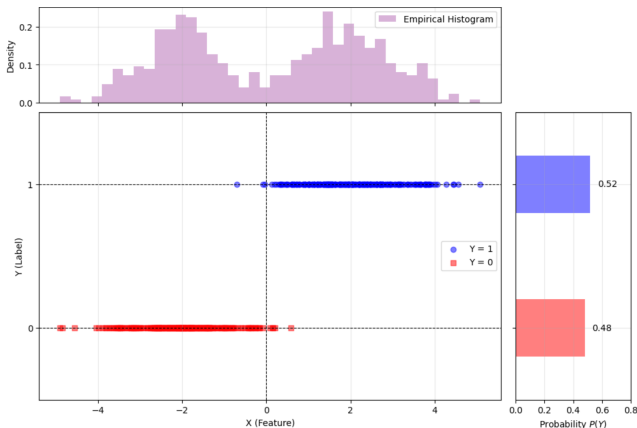
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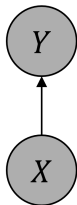
Discriminative vs. Generative Models



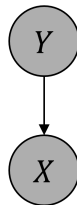
- ▶ Given data, there are multiple ways to describe the dependency structure.
- ▶ For example, let X be an explanatory variable and Y be a binary response.
- ▶ Q: How can we construct a probabilistic model for $(X, Y)^T$ and learn $p(y | x)$?^a

^aThis task is referred to as *binary classification*.

Discriminative vs. Generative Models



(a) Discriminative Model



(b) Generative Model

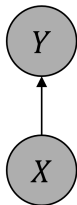
► We may consider two approaches:

1. **Discriminative Model**: Models only the conditional distribution $p(y|x)$.
2. **Generative Model**: Models the joint distribution

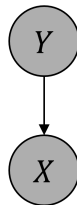
$$f(x, y) = p(y)f(x | y), \tag{1}$$

typically by modelling $p(y)$ and $f(x|y)$ separately.

Discriminative Model



(a) Discriminative Model



(b) Generative Model

- **Discriminative Model**: We consider *logistic regression*. First, we may consider:

$$X \sim f(x) \quad \text{and} \quad Y \mid X = x \sim \text{Ber}(\mu_\theta(x)), \quad (2)$$

where $f(x)$ does not depend on the parameter θ , and $\mu_\theta(x)$ is a parameterized conditional success probability designed to model the dependence of Y on X .

Discriminative Model

- ▶ In this case, for a given x , the conditional log-PMF is expressed as:

$$\begin{aligned}\log p_{\theta}(y | x) &= \log \text{Ber}(y; \mu_{\theta}(x)) \\ &= \log \left(\mu_{\theta}(x)^y (1 - \mu_{\theta}(x))^{1-y} \right) \\ &= y \log \frac{\mu_{\theta}(x)}{1 - \mu_{\theta}(x)} + \log(1 - \mu_{\theta}(x)).\end{aligned}\tag{3}$$

- ▶ Note that $\log p_{\theta}(y | x)$ is a linear function of y with the coefficient:

$$\eta_{\theta}(x) = \log \frac{\mu_{\theta}(x)}{1 - \mu_{\theta}(x)},^1\tag{4}$$

which is the *logit* (or *log-odds*) of success given x .

¹It is formally referred to as the *natural parameter* in the context of the *exponential family*.

Discriminative Model

- ▶ Here, letting $\text{sig}(u) := 1/(1 + \exp(-u))$ denote the *sigmoid function* (or *logistic function*), we have:

$$\mu_{\theta}(x) = \text{sig}(\eta_{\theta}(x)) = \frac{1}{1 + \exp(-\eta_{\theta}(x))}.^2 \quad (5)$$

- ▶ For the choice of $\eta_{\theta}(\cdot)$, we consider the *logistic model* where the logit $\eta_{\theta}(x)$ is a linear function of x :

$$\left\{ \eta_{\theta}(x) = \beta_0 + \beta_1 x \mid \theta = (\beta_0, \beta_1)^T \in \mathbb{R}^2 \right\}. \quad (6)$$

- ▶ Finding the MLE within this model class leads to *logistic regression*.

²In a similar manner, this formulation can be extended to multi-class classification using the *softmax function*.

Discriminative Model

- ▶ Let's formulate the likelihood maximization. The (joint) log-likelihood function can be expressed as:

$$\begin{aligned}\ell(\theta) &= \sum_{i=1}^n \log f_{\theta}(x_i, y_i) \\ &= \sum_{i=1}^n \log f(x_i) + \sum_{i=1}^n \left(y_i \eta_{\theta}(x_i) + \log(1 - \text{sig}(\eta_{\theta}(x_i))) \right).\end{aligned}\tag{7}$$

- ▶ Differentiating both sides with respect to θ yields:

$$\nabla_{\theta} \ell(\theta) = \sum_{i=1}^n (y_i - \text{sig}(\eta_{\theta}(x_i))) \nabla_{\theta} \eta_{\theta}(x_i).\tag{8}$$

Discriminative Model

- ▶ Differentiating both sides once more yields the *Hessian matrix*:

$$\nabla_{\theta}^2 \ell(\theta) = - \sum_{i=1}^n \text{sig}(\eta_{\theta}(x_i)) (1 - \text{sig}(\eta_{\theta}(x_i))) \nabla_{\theta} \eta_{\theta}(x_i) \nabla_{\theta} \eta_{\theta}(x_i)^T. \quad (9)$$

- ▶ For the logistic model class where $\eta_{\theta}(x) = \beta_0 + \beta_1 x$, we have:

$$\nabla_{\theta} \eta_{\theta}(x_i) = (1, x_i)^T. \quad (10)$$

Consequently, the Hessian $\nabla_{\theta}^2 \ell(\theta)$ becomes strictly negative definite provided that the sample variance satisfies $S_X^2 > 0$.

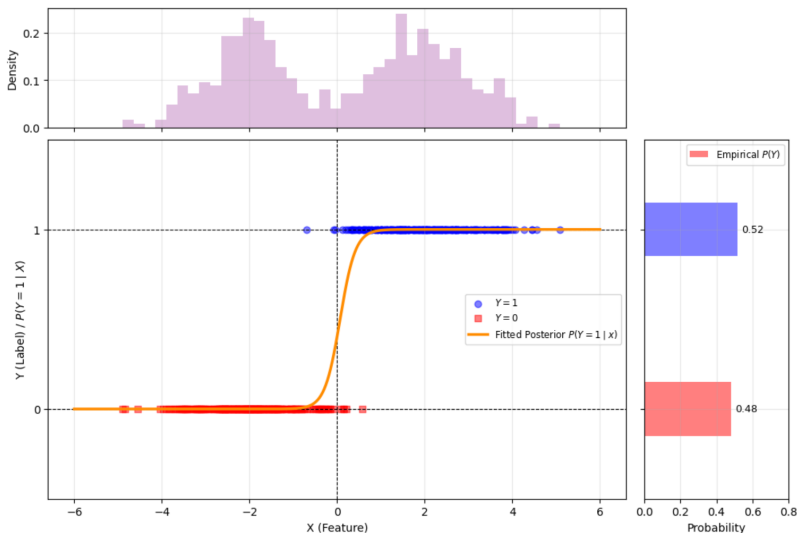
Discriminative Model

- ▶ Thus, $\ell(\theta)$ is strictly concave, and it is sufficient to solve the *score equation*:

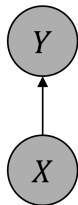
$$\nabla_{\theta} \ell(\theta) = 0, \quad \text{or equivalently,}$$
$$\sum_{i=1}^n (y_i - \mu_{\theta}(x_i)) \begin{pmatrix} 1 \\ x_i \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (11)$$

- ▶ We can apply Newton's method, an iterative optimization algorithm, to numerically approximate the MLE.
- ▶ Introducing $\hat{p}_i := \mu_{\theta}(x_i) = P_{\theta}(Y = 1 \mid X = x_i)$, we have:

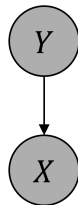
Discriminative Model



Generative Model



(a) Discriminative Model



(b) Generative Model

- **Generative Model**: We consider *linear discriminant analysis (LDA)*. First, we model the joint distribution directly:

$$Y \sim \text{Ber}(p) \quad \text{and} \quad X \mid Y = y \sim \mathcal{N}(\mu_y, \sigma^2), \quad (12)$$

where $p \in (0, 1)$ is the class prior, and the conditional distribution of X given Y is modeled as a class-specific Gaussian with a shared variance σ^2 .

Generative Model

- ▶ Let $\theta = (p, \mu_0, \mu_1, \sigma^2)^T$ be the parameter vector. Then, the joint density-PMF can be expressed as:

$$f_{\theta}(x, y) = \underbrace{\left(p^y (1-p)^{1-y} \right)}_{=p_{\theta}(y)} \underbrace{\left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(x - \mu_y)^2}{2\sigma^2} \right) \right)}_{=f_{\theta}(x|y)}. \quad (13)$$

- ▶ Consequently, by Bayes' rule, the posterior probabilities are given by:

$$P_{\theta}(Y = 1 \mid X = x) = \frac{f_{\theta}(x \mid y = 1)p_{\theta}(y = 1)}{f_{\theta}(x \mid y = 1)p_{\theta}(y = 1) + f_{\theta}(x \mid y = 0)p_{\theta}(y = 0)} \quad (14)$$

and

$$P_{\theta}(Y = 0 \mid X = x) = \frac{f_{\theta}(x \mid y = 0)p_{\theta}(y = 0)}{f_{\theta}(x \mid y = 1)p_{\theta}(y = 1) + f_{\theta}(x \mid y = 0)p_{\theta}(y = 0)}. \quad (15)$$

Generative Model

- ▶ Thus, given θ , the corresponding logit can be expressed as:

$$\eta_{\theta}(x) := \log \frac{P_{\theta}(Y = 1 \mid X = x)}{P_{\theta}(Y = 0 \mid X = x)}. \quad (16)$$

Generative Model

- ▶ Consequently, the decision boundary of this model, defined as the set of points where $\eta_{\theta}(x) = 0$, is strictly *linear* with respect to x :

$$\beta_0 + \beta_1 x = 0, \quad (17)$$

where we define the mapped parameters as:

$$\beta_0 := \log \frac{p}{1-p} - \frac{\mu_1^2 - \mu_0^2}{2\sigma^2} \quad \text{and} \quad \beta_1 := \frac{\mu_1 - \mu_0}{\sigma^2}. \quad (18)$$

- ▶ This demonstrates that the LDA naturally induces a logistic regression structure. We can thus use its MLE to calculate the predicted posterior probabilities.
- ▶ Conversely, the LDA can represent any (non-trivial) logistic model. When $\beta_1 \neq 0$,

$$(p, \mu_0, \mu_1, \sigma^2)^T = \left(0.5, -\frac{\beta_0}{\beta_1} - \frac{\beta_1}{2}, -\frac{\beta_0}{\beta_1} + \frac{\beta_1}{2}, 1 \right)^T \quad (19)$$

recovers the corresponding logistic model with $(\beta_0, \beta_1)^T$.

Generative Model

- Let's formulate the likelihood maximization. The log-likelihood function is given by:

$$\begin{aligned}\ell(\theta) &= \sum_{i=1}^n \log \left(p^{y_i} (1-p)^{1-y_i} \right) + \sum_{i=1}^n \log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(x_i - \mu_{y_i})^2}{2\sigma^2} \right) \right) \\ &= \sum_{i=1}^n \left(y_i \log p + (1-y_i) \log(1-p) - \frac{1}{2} \log(2\pi\sigma^2) - \frac{(x_i - \mu_{y_i})^2}{2\sigma^2} \right).\end{aligned}\tag{20}$$

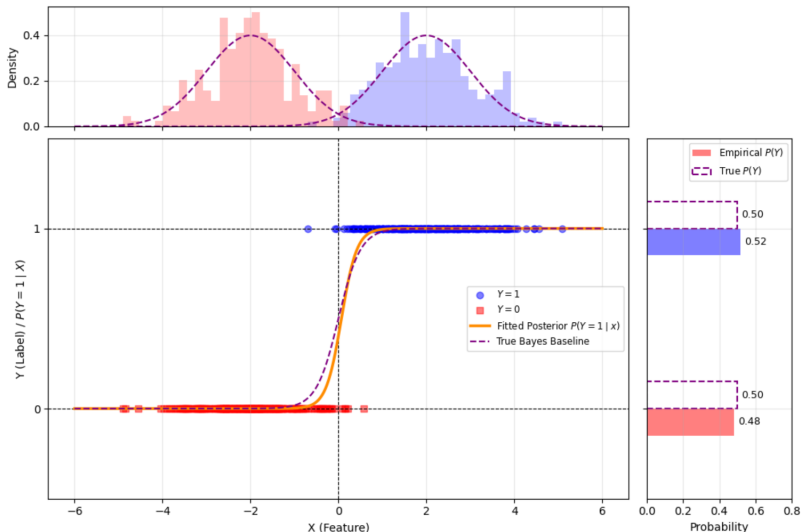
- Unlike logistic regression, $\ell(\theta)$ can be optimized analytically in closed form:

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n y_i, \quad \hat{\mu}_1 = \frac{\sum_{i=1}^n y_i x_i}{\sum_{i=1}^n y_i}, \quad \hat{\mu}_0 = \frac{\sum_{i=1}^n (1-y_i) x_i}{\sum_{i=1}^n (1-y_i)}, \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu}_{y_i})^2.\tag{21}$$

Generative Model

$$\blacktriangleright \ell(\theta) = \sum_{i=1}^n \left(y_i \log p + (1 - y_i) \log(1 - p) - \frac{1}{2} \log(2\pi\sigma^2) - \frac{(x_i - \mu_{y_i})^2}{2\sigma^2} \right).$$

Generative Model



Generative Model: Synthetic Data Generation

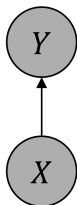
- Given the LDA generative model $f_{\theta}(x, y) = p_{\theta}(y)f_{\theta}(x | y)$, we can *generate* synthetic data as follows:
1. Draw a new label $y_{\text{new}} \sim \text{Ber}(p)$.
 2. Generate a uniform realization $u \sim \mathcal{U}(0, 1)$.
 3. Calculate x_{new} :

$$x_{\text{new}} = \mu_{y_{\text{new}}} + \sigma \Phi^{-1}(u). \quad (22)$$

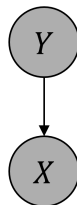
- More generally, for any class-conditional distribution with a known CDF $F_{X|Y=y}(x)$, we can draw $u \sim \mathcal{U}(0, 1)$ and compute:

$$x_{\text{new}} = F_{X|Y=y_{\text{new}}}^{-1}(u). \quad (23)$$

Discriminative vs. Generative Models



(a) Discriminative Model



(b) Generative Model

- Q: What would be the pros and cons of the two approaches?

(a) **Logistic Regression**: Directly models the posterior probability:

$$P_{\theta}(Y = 1 | X = x) = \text{sig}(\beta_0 + \beta_1 x). \quad (24)$$

(b) **Linear Discriminant Analysis**: Models the entire data-generating process:

$$f_{\theta}(x, y) = p_{\theta}(y)f_{\theta}(x | y), \quad (25)$$

where $Y \sim \text{Ber}(p)$, $X | Y = y \sim \mathcal{N}(\mu_y, \sigma^2)$, and $\theta = (p, \mu_0, \mu_1, \sigma^2)^T$.

Discriminative vs. Generative Models

- ▶ In the previous toy data, the true data-generating mechanism was as follows:

$$Y \sim \text{Ber}(p) \quad \text{and} \quad X \mid Y = y \sim \mathcal{N}(\mu_y, \sigma^2) \quad (26)$$

with $(p, \mu_0, \mu_1, \sigma^2)^T = (0.5, -2, 2, 1)^T$, implying $(\beta_0, \beta_1)^T = (0, 4)^T$. Both models correctly specify the true linear logit structure.

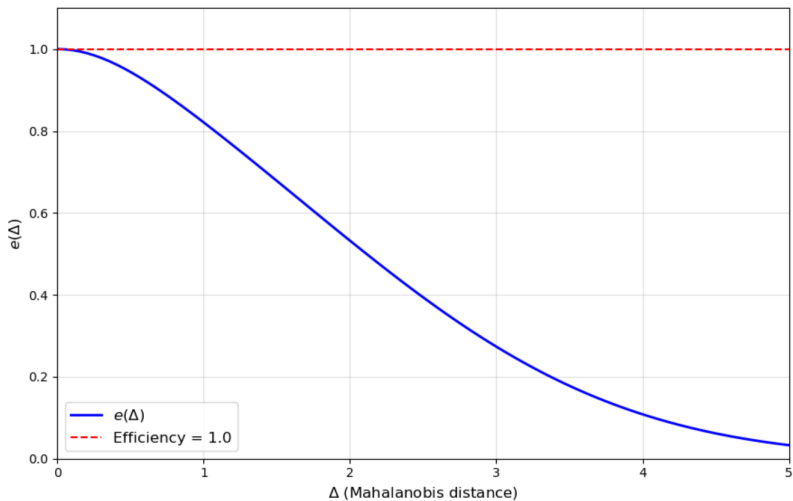
- ▶ When $p = 0.5$, the relative asymptotic efficiency for β_1 is expressed as (Efron, 1975):

$$e(\Delta) := \lim_{n \rightarrow \infty} \frac{\text{Var}(\hat{\beta}_{1,\text{LDA}})}{\text{Var}(\hat{\beta}_{1,\text{LR}})} = \frac{4 + \Delta^2}{2} \int_{-\infty}^{\infty} z^2 \frac{\phi\left(z - \frac{\Delta}{2}\right) \phi\left(z + \frac{\Delta}{2}\right)}{\phi\left(z - \frac{\Delta}{2}\right) + \phi\left(z + \frac{\Delta}{2}\right)} dz, \quad (27)$$

where $\Delta = \frac{|\mu_1 - \mu_0|}{\sigma}$ represents the *Mahalanobis distance*.

- ▶ In our specific setup ($\Delta = 4$), the efficiency for the slope is $e(4) \approx 0.195$.

Discriminative vs. Generative Models



Discriminative vs. Generative Models

- ▶ Next, suppose that the true underlying data-generating mechanism follows:

$$Y \sim \text{Ber}(0.5), \quad X \mid Y = 0 \sim -2 + t(2), \quad \text{and} \quad X \mid Y = 1 \sim 2 + t(2). \quad (28)$$

Here, $t(2)$ denotes a Student's t -distribution with 2 degrees of freedom, whose PDF is given by:

$$f(x) = \frac{1}{2\sqrt{2}} \left(1 + \frac{x^2}{2} \right)^{-3/2}. \quad (29)$$

Due to this structural form, it exhibits a much higher probability of observing extreme values (heavy tails) than a Gaussian distribution.

- ▶ We then investigate the consequences of naively applying both logistic regression and LDA, completely ignoring this model misspecification.

Discriminative vs. Generative Models

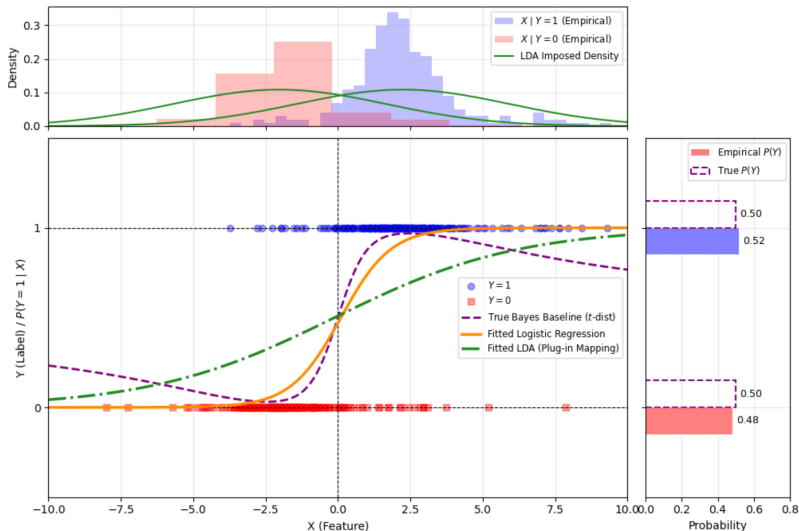
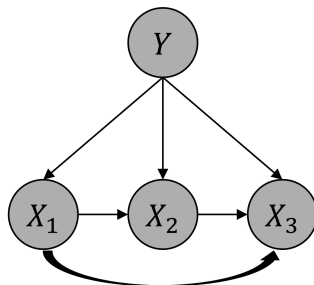


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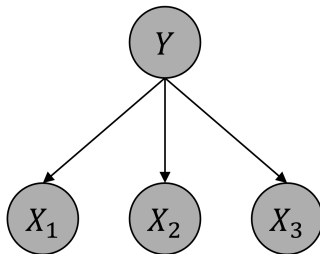
Multivariate Generative Model



- ▶ Multivariate data often exhibits complex dependency structures.
- ▶ The generative model requires learning $f_{\theta}(\mathbf{x} \mid y)$. For example, when $d = 3$:

$$\begin{aligned} f_{\theta}(\mathbf{x}, y) &= p_{\theta}(y) f_{\theta}(\mathbf{x} \mid y) \\ &= p_{\theta}(y) \left(f_{\theta}(x_1 \mid y) f_{\theta}(x_2 \mid y, x_1) f_{\theta}(x_3 \mid y, x_1, x_2) \right). \end{aligned} \tag{30}$$

Multivariate Generative Model: Naive Bayes



- ▶ *Naive Bayes* is a multivariate generative model assuming complete conditional independence:

$$f_{\theta}(\mathbf{x} \mid y) = \prod_{j=1}^d f_{\theta}(x_j \mid y). \quad (31)$$

Multivariate Generative Model: Naive Bayes

- ▶ The assumption of conditional independence simplifies the logit structure:

$$\begin{aligned}\eta_{\theta}(\mathbf{x}) &= \log \frac{P_{\theta}(Y = 1 \mid \mathbf{X} = \mathbf{x})}{P_{\theta}(Y = 0 \mid \mathbf{X} = \mathbf{x})} \\ &= \log \left(\frac{f_{\theta}(\mathbf{x} \mid y = 1)p_{\theta}(y = 1)}{f_{\theta}(\mathbf{x} \mid y = 0)p_{\theta}(y = 0)} \right) \\ &= \sum_{j=1}^d \log \frac{f_{\theta}(x_j \mid y = 1)}{f_{\theta}(x_j \mid y = 0)} + \log \frac{p_{\theta}(y = 1)}{p_{\theta}(y = 0)}.\end{aligned}\tag{32}$$

- ▶ The problem of learning $\log \frac{f_{\theta}(\mathbf{x}|y=1)}{f_{\theta}(\mathbf{x}|y=0)}$ reduces to learning d univariate logits.
- ▶ For example, if $X_j \mid Y = y \sim \mathcal{N}(\mu_{yj}, \sigma_j^2)$, the logit becomes linear:

$$\eta_{\theta}(\mathbf{x}) = \sum_{j=1}^d \left(\frac{\mu_{1j} - \mu_{0j}}{\sigma_j^2} x_j + \frac{\mu_{0j}^2 - \mu_{1j}^2}{2\sigma_j^2} \right) + \log \frac{p_{\theta}(y = 1)}{p_{\theta}(y = 0)}.\tag{33}$$

Multivariate Generative Model: Multivariate LDA

- ▶ Unlike Naive Bayes, multivariate LDA models the full dependency structure:

$$\mathbf{X} \mid Y = y \sim \mathcal{N}(\boldsymbol{\mu}_y, \Sigma). \quad (34)$$

- ▶ The covariance matrix Σ captures dependencies between features and is shared across all classes.
- ▶ The conditional PDF is given by:

$$f_{\theta}(\mathbf{x} \mid y) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_y)^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_y) \right). \quad (35)$$

where $\theta = (\boldsymbol{\mu}_0^T, \boldsymbol{\mu}_1^T, \text{vec}(\Sigma))^T$ is the parameter set containing $O(d^2)$ parameters.

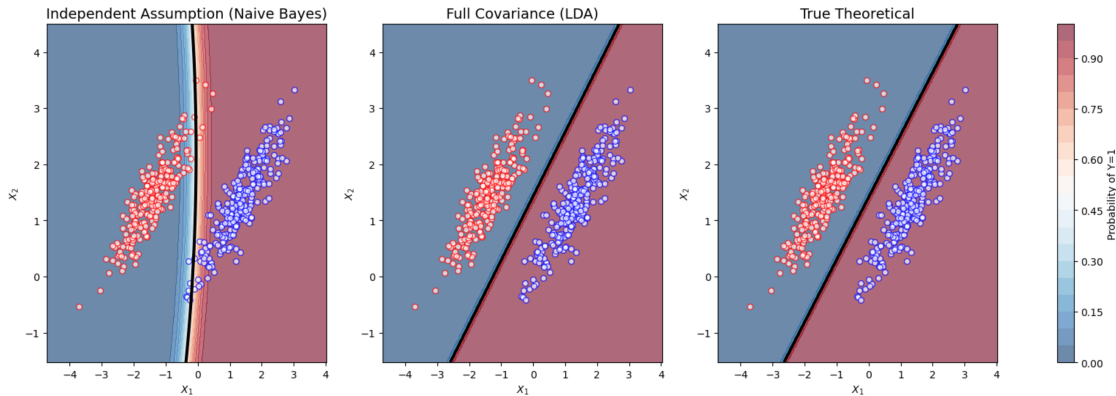
Multivariate Generative Model: Multivariate LDA

- ▶ The logit $\eta_{\theta}(\mathbf{x})$ can be expressed as:

$$\begin{aligned}\eta_{\theta}(\mathbf{x}) &= \log \frac{f_{\theta}(\mathbf{x} \mid y = 1)p_{\theta}(y = 1)}{f_{\theta}(\mathbf{x} \mid y = 0)p_{\theta}(y = 0)} \\ &= (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \boldsymbol{\Sigma}^{-1} \left(\mathbf{x} - \frac{\boldsymbol{\mu}_1 + \boldsymbol{\mu}_0}{2} \right) + \log \frac{p_{\theta}(y = 1)}{p_{\theta}(y = 0)}.\end{aligned}\tag{36}$$

- ▶ The logit remains linear. However, $\boldsymbol{\Sigma}^{-1}$ explicitly accounts for cross-feature dependencies when defining the weight vector $\mathbf{w} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$.
- ▶ When $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}_d$, multivariate LDA reduces to Gaussian naive Bayes.

Multivariate Generative Model



Curse of Dimensionality

- ▶ As d grows much faster than the sample size n ($d/n \rightarrow \infty$), multivariate LDA performs no better than random guessing (Bickel and Levina, 2004).
- ▶ When $P(Y = 1) = 0.5$, the plug-in classification rule based on multivariate LDA is formulated as:

$$\hat{\delta}_{\text{LDA}}(\mathbf{X}) = I \left((\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0)^T \hat{\boldsymbol{\Sigma}}^{-1} \left(\mathbf{X} - \frac{\hat{\boldsymbol{\mu}}_1 + \hat{\boldsymbol{\mu}}_0}{2} \right) > 0 \right) \quad (37)$$

where $\hat{\boldsymbol{\mu}}_y$ and $\hat{\boldsymbol{\Sigma}}$ are the MLEs.

- ▶ Note that a random guessing classifier defined as:

$$\delta_{\text{Random}}(\mathbf{X}) = I(U < 0.5), \quad \text{where } U \sim \mathcal{U}(0, 1), \quad (38)$$

yields a misclassification error rate of $P_{\theta}(\delta_{\text{Random}}(\mathbf{X}) \neq Y) = 0.5$.

Curse of Dimensionality

- ▶ The misclassification error rate of multivariate LDA satisfies:

$$P_{\theta}(\hat{\delta}_{\text{LDA}}(\mathbf{X}) \neq Y) \rightarrow 0.5 \quad (39)$$

when $d/n \rightarrow \infty$, even if the true covariance matrix Σ is well-conditioned with bounded eigenvalues (e.g., $\Sigma = I_d$) (Bickel and Levina, 2004).

- ▶ The root cause of this curse of dimensionality is the excessive number of parameters to estimate.
- ▶ When $d \gg n$, the sample covariance matrix $\hat{\Sigma}$ fails to estimate the $O(d^2)$ elements of Σ . Specifically, $\hat{\Sigma}$ becomes singular, making it challenging to compute $\hat{\Sigma}^{-1}$.

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1. Discriminative vs. Generative Models

2. Curse of Dimensionality

3. Dimensionality Reduction

Linear Discriminant Analysis

- ▶ *Dimension reduction* is the process of identifying low-dimensional features that capture the essential information in the data and transforming the data accordingly.
- ▶ For example, the LDA classification decision,

$$\hat{\delta}_{\text{LDA}}(\mathbf{X}) = I \left((\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0)^T \hat{\boldsymbol{\Sigma}}^{-1} \left(\mathbf{X} - \frac{\hat{\boldsymbol{\mu}}_1 + \hat{\boldsymbol{\mu}}_0}{2} \right) > 0 \right), \quad (40)$$

is based on the scalar projection:

$$\underbrace{\left(\hat{\boldsymbol{\Sigma}}^{-1} (\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0) \right)^T}_{=:\mathbf{w}^T} \left(\mathbf{X} - \frac{\hat{\boldsymbol{\mu}}_1 + \hat{\boldsymbol{\mu}}_0}{2} \right). \quad (41)$$

- ▶ In this context, LDA can be viewed as a supervised dimension reduction method that utilizes the class label Y .

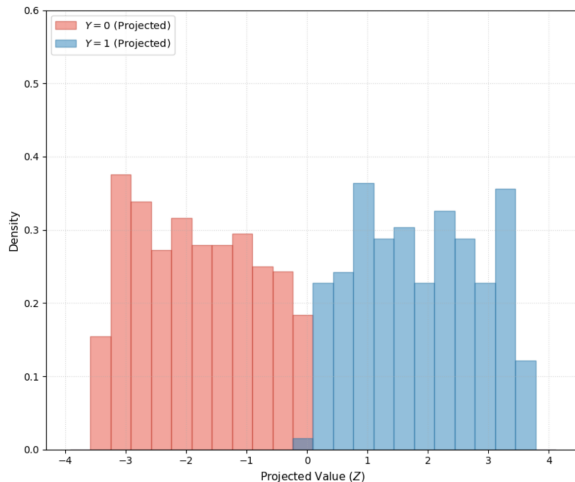
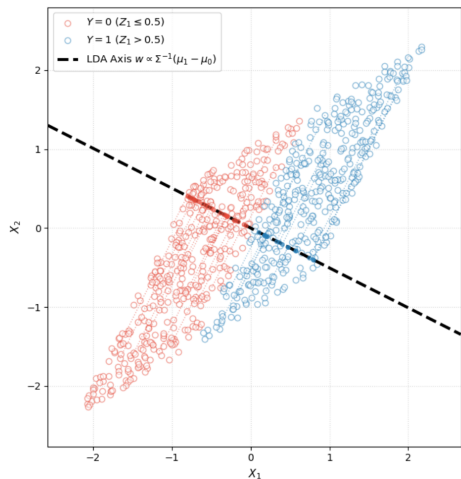
Linear Discriminant Analysis

- ▶ Even when the conditional Gaussian assumption is violated, the LDA projection axis can still be justified as the solution to:

$$\hat{\Sigma}^{-1}(\hat{\mu}_1 - \hat{\mu}_0) \propto \arg \max_{\mathbf{w}: \|\mathbf{w}\|=1} \left| \widehat{\text{Var}}^{-1/2}[\mathbf{w}^T \mathbf{X}] \left(\hat{\mathbb{E}}[\mathbf{w}^T \mathbf{X} \mid Y=1] - \hat{\mathbb{E}}[\mathbf{w}^T \mathbf{X} \mid Y=0] \right) \right|. \quad (42)$$

- ▶ That is, the LDA projection axis maximizes the Mahalanobis distance between the two class centers.

Linear Discriminant Analysis



Principal Component Analysis

- ▶ While LDA effectively identifies the axis that discriminates between the two classes, labeled data are typically scarce in practice.
- ▶ Principal Component Analysis (PCA) is a popular unsupervised dimension reduction method that does not utilize class labels.
- ▶ PCA first finds the direction preserving the variability in the data as much as possible:

$$\mathbf{w}_1 = \arg \max_{\mathbf{w}_1: \|\mathbf{w}_1\|=1} \widehat{\text{Var}}[\mathbf{w}_1^T \mathbf{X}] \quad (43)$$

- ▶ Consequently, the r -th principal axis is sequentially obtained by maximizing the remaining variance, subject to being orthogonal to all previously defined axes:

$$\mathbf{w}_j = \arg \max_{\substack{\mathbf{w}_j: \|\mathbf{w}_j\|=1, \\ \mathbf{w}_j^T \mathbf{w}_k=0, \forall k < j}} \widehat{\text{Var}}[\mathbf{w}_j^T \mathbf{X}] \quad (44)$$

Principal Component Analysis

- ▶ Since $\widehat{\text{Var}}[\mathbf{w}_j^T \mathbf{X}] = \mathbf{w}_j^T \hat{\Sigma} \mathbf{w}_j$, the sample covariance matrix $\hat{\Sigma}$ fully characterizes all of the principal components (PCs).
- ▶ The PCA axes correspond to the eigenvectors of $\hat{\Sigma}$, where the associated eigenvalues λ_j directly quantify the explained variance along each component:

$$\hat{\Sigma} \mathbf{w}_j = \lambda_j \mathbf{w}_j \quad \text{where} \quad \lambda_j = \widehat{\text{Var}}[\mathbf{w}_j^T \mathbf{X}] \quad (45)$$

where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$.

- ▶ The total sample variance, representing the entire variability in $\mathbf{X}$, is preserved under this orthogonal transformation:

$$\sum_{j=1}^d \widehat{\text{Var}}[X_j] = \text{tr}(\hat{\Sigma}) = \sum_{j=1}^d \lambda_j = \sum_{j=1}^d \widehat{\text{Var}}[\mathbf{w}_j^T \mathbf{X}] \quad (46)$$

Principal Component Analysis

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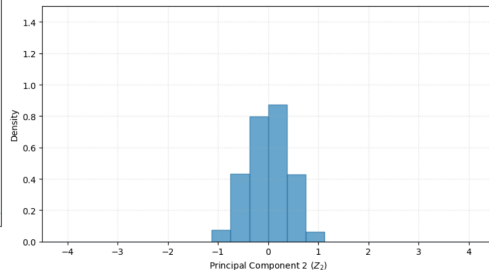
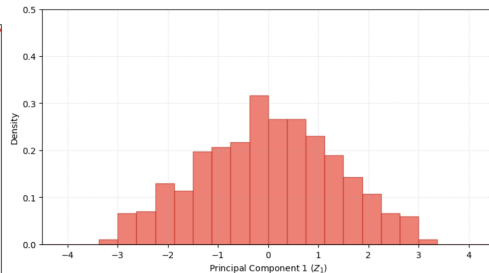
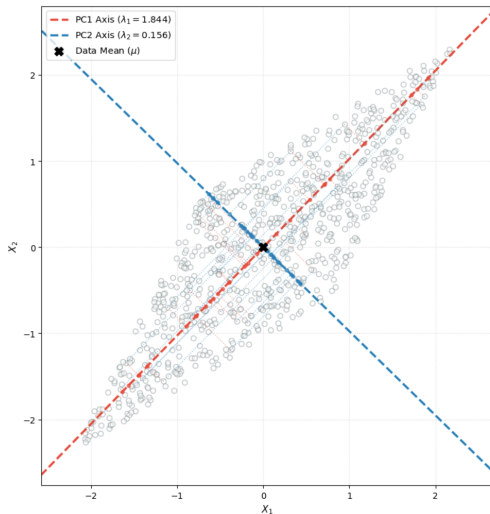
$$\hat{\Sigma} \mathbf{w}_j = \lambda_j \mathbf{w}_j \quad \text{where} \quad \lambda_j = \widehat{\text{Var}}[\mathbf{w}_j^T \mathbf{X}] \quad (47)$$

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Principal Component Analysis



Principal Component Analysis

- ▶ We can reconstruct the data vector $\mathbf{X}$ using the top- k PCs through a mirrored encoding-decoding mapping:

$$\mathbf{X} \xrightarrow{\text{Embedding}} W^T(\mathbf{X} - \hat{\boldsymbol{\mu}}) \xrightarrow{\text{Reconstruction}} \hat{\boldsymbol{\mu}} + WW^T(\mathbf{X} - \hat{\boldsymbol{\mu}}) \quad (49)$$

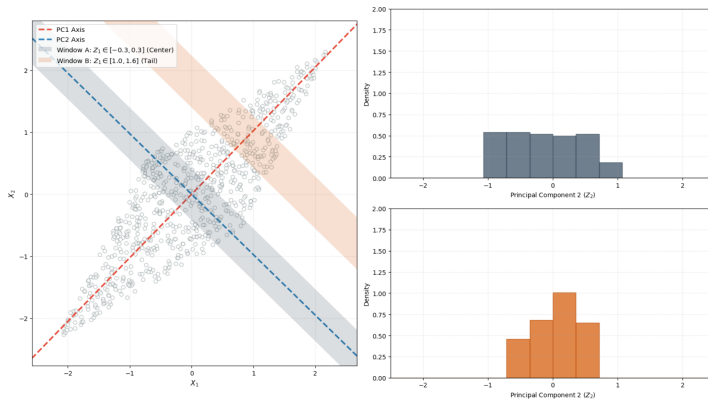
where $W = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k]$ represents the $d \times k$ projection matrix.

- ▶ The mean squared error (called *reconstruction error*) under this subspace approximation can be explicitly quantified as:

$$n^{-1} \sum_{i=1}^n \left\| \mathbf{x}_i - \left(\hat{\boldsymbol{\mu}} + WW^T(\mathbf{x}_i - \hat{\boldsymbol{\mu}}) \right) \right\|^2 = \text{tr}(\hat{\boldsymbol{\Sigma}}) - \sum_{j=1}^k \widehat{\text{Var}}[\mathbf{w}_j^T \mathbf{X}] \quad (50)$$

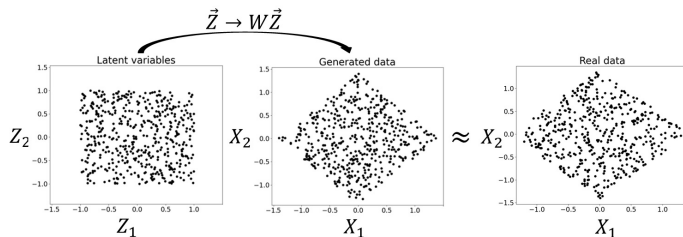
- ▶ Thus, the PCA axes can be interpreted as the minimizers of the reconstruction error under the orthonormality constraint.

Independent Component Analysis



- ▶ Although PCA identifies orthogonal components, orthogonality or uncorrelatedness alone does not imply statistical independence.

Independent Component Analysis



- ▶ Independent Component Analysis (ICA) (Hyvärinen and Oja, 2000) identifies latent sources and utilizes them to achieve statistically independent dimension reduction.
- ▶ ICA assumes a linear generative structure: $\mathbf{X} = W\mathbf{Z}$, where $W \in \mathbb{R}^{d \times k}$ and $W^T W = I_k$. Here, $\mathbf{Z}$ denotes latent variables following a known non-Gaussian distribution (e.g., $\mathcal{U}([0, 1]^2)$ or $\mathcal{U}([-1, 1]^2)$).
- ▶ Assuming all underlying sources are mutually independent, the latent variables can be explicitly inferred via $\mathbf{Z} = W^T \mathbf{X}$.

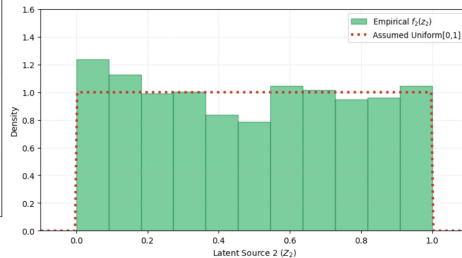
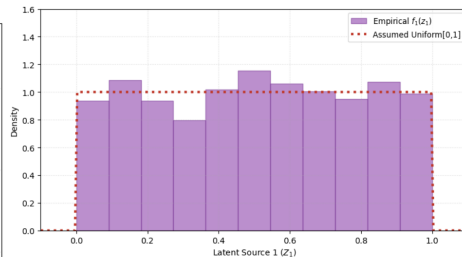
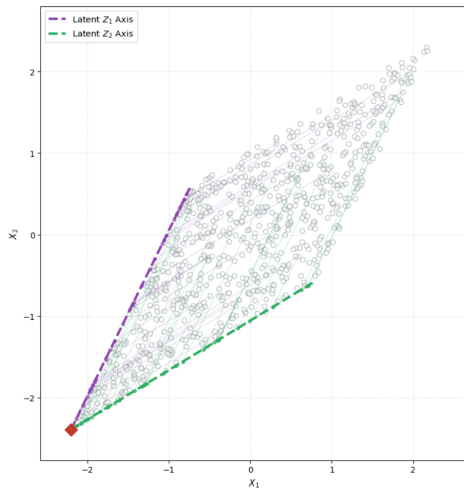
Independent Component Analysis

- ▶ There are several criteria for learning this linear generative structure, including kurtosis maximization, mutual information minimization, and maximum likelihood estimation.
- ▶ Focusing on the likelihood-maximization approach,³ the change of variables formula yields $p_{\theta}(\mathbf{x}) = p_{\mathbf{Z}}(\mathbf{z} = W^T \mathbf{x}) \cdot |\det W^T|$.
- ▶ Since W is orthogonal ($|\det W^T| = 1$), the log-likelihood $l_n(\theta)$ simplifies via strict statistical independence to:

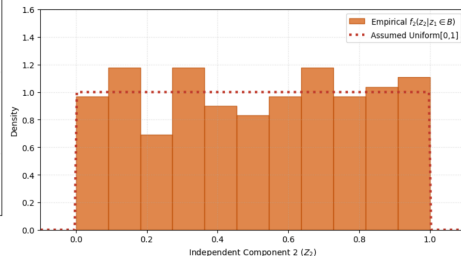
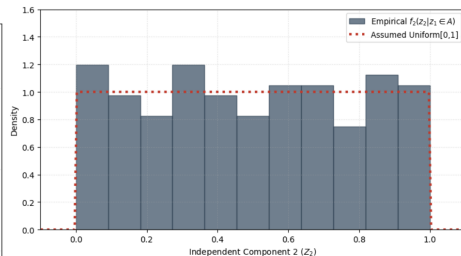
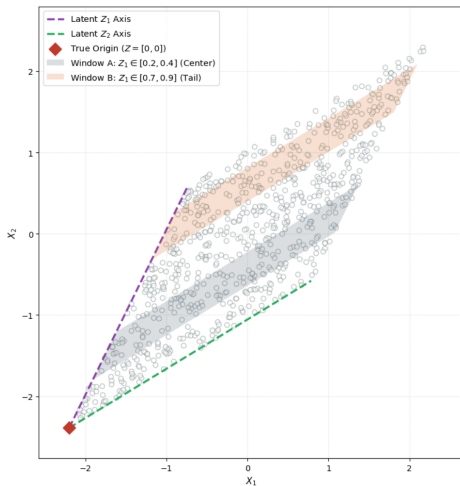
$$l_n(\theta) = \sum_{i=1}^n \log p_{\mathbf{Z}}(W^T \mathbf{x}_i) = \sum_{i=1}^n \sum_{j=1}^d \log p_{Z_j} \left((W^T \mathbf{x}_i)_j \right) \quad (51)$$

³Although ICA is generally applicable when $k < d$, for simplicity, we assume $k = d$ so that $W^T = W^{-1}$.

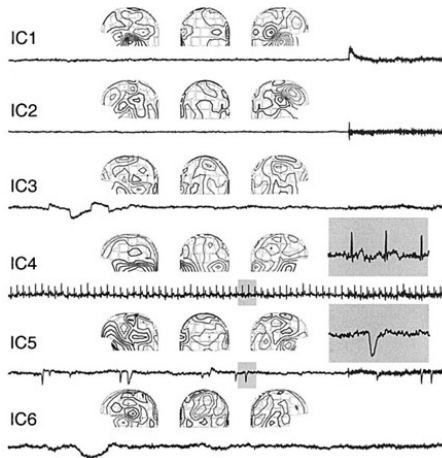
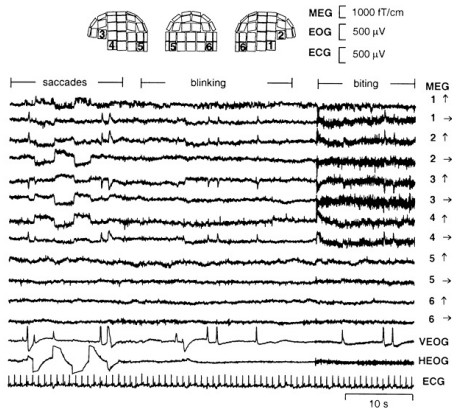
Independent Component Analysis



Independent Component Analysis

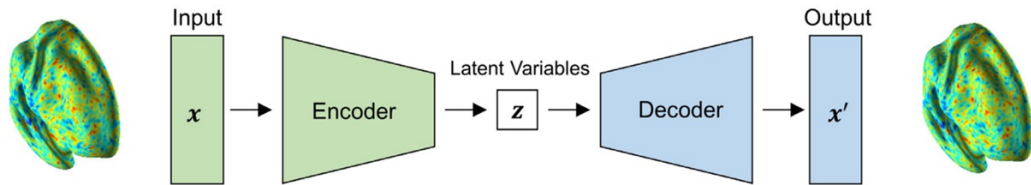


Independent Component Analysis



Figures are edited from Hyvärinen and Oja (2000).

Nonlinear Methods: Autoencoders



- ▶ As a non-linear extension of PCA, we can generalize the framework using an autoencoder (Kramer, 1991; Plaut, 2018).
- ▶ Autoencoders (AEs) (Bengio et al., 2006) consist of pairs of encoders and decoders that efficiently reduce the dimensionality of data.
- ▶ The prefix 'auto' is used because they autonomously learn to encode data in an unsupervised manner.

Images are from Kim et al. (2021).

Nonlinear Methods: Autoencoders

- Specifically, we may consider a flexible encoding function $f_{\text{enc}}(\cdot)$ and a linear decoder W :

$$\mathbf{X} \xrightarrow{\text{Embedding}} f_{\text{enc}}(\mathbf{X}) \xrightarrow{\text{Reconstruction}} \mathbf{b} + Wf_{\text{enc}}(\mathbf{X}) \quad (52)$$

- To understand its foundation, consider minimizing the reconstruction error over all possible encoding results $f_{\text{enc}}(\mathbf{x}_1), \dots, f_{\text{enc}}(\mathbf{x}_n) \in \mathbb{R}^k$ subject to $W^T W = I_k$:

$$n^{-1} \sum_{i=1}^n \|\mathbf{x}_i - (\mathbf{b} + Wf_{\text{enc}}(\mathbf{x}_i))\|^2 \geq \text{tr}(\hat{\Sigma}) - \sum_{j=1}^k \widehat{\text{Var}}[\mathbf{w}_j^T \mathbf{X}] \quad (53)$$

- It is known that the lower bound is achieved if and only if:

$$\mathbf{b} + Wf_{\text{enc}}(\mathbf{x}_i) = \hat{\boldsymbol{\mu}} + W^T(\mathbf{x}_i - \hat{\boldsymbol{\mu}})$$

This implies that the first k PCs are the optimal choices.

- See Section 14.5.1 in Hastie (2009) for further theoretical details.

Nonlinear Methods: Variational Autoencoders

- ▶ Variational autoencoders (VAEs) (Kingma and Welling, 2014) can be understood as a non-linear extension of Independent Component Analysis (ICA) (Khemakhem et al., 2020).
- ▶ A more rigorous and detailed discussion on variational autoencoders will be presented in a later chapter.

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